

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Data Structures and Algorithms [CSS_5114]

Marks: 50

Duration: 180 mins.

PART-B

Answer all the questions.

- 1) Consider the following sequence of insertions into an AVL tree: (5)
20, 15, 10, 25, 30, 5, 35, 40, 45, 50

- A)
 - Draw the AVL tree and perform rotations where necessary.

Explain the balance factor at each step and describe when rotations are required.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Implement **two stacks in an array**. Perform the following (3)
operations:

- **Push:** 12, 24, 36, 48, 60, 72
- **Pop** two elements from the first stack, and then push

84 and **96** into the second stack.

Pop from both stacks and show the results.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Define an array of **10 integers**: (2)
arr[10] = {3, 6, 9, 12, 15, 18, 21, 24, 27, 30}
Use pointers to calculate the sum and average of all elements.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2) Convert the following infix expression to **prefix**: (5)
A + B * (C - D) / E

- A) Show the step-by-step conversion process using a stack.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Evaluate the following postfix expression: (3)
5 6 + 8 2 / * 7 +
Write a function to do the same and illustrate with suitable diagrams.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Define a structure for a Student that includes name, roll number, and CGPA. Inside this structure, (2)
define another structure for Address (with fields house number, city, and pincode). Write a program
to store details of 3 students and update the city of the second student using pointers to structures.

- 3) Sort the following array using **Heap Sort**:
80, 23, 45, 67, 12, 55, 99, 15, 30, 73 (5)

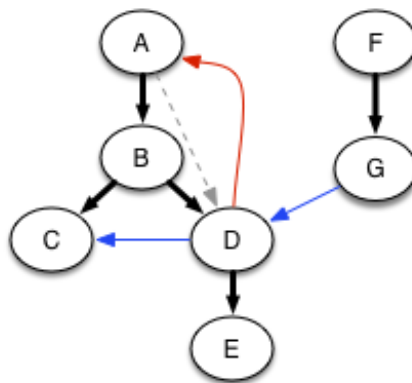
- A)
 - Explain each step involved in heap sort.

Develop the code for the same.

- B) Write a recursive function to compute the **nth Fibonacci number**. (3)

- The Fibonacci sequence is defined as:
 $F(0) = 0, F(1) = 1, F(n) = F(n-1) + F(n-2) \forall n > 1$
- Implement the function recursively to find **F(10)** and show the sequence of recursive calls.
- Discuss the time complexity of your recursive solution.

- C) Based on the DFS traversal of the given graph, classify each edge as **Tree edge**, **Back edge**, **Forward edge**, or **Cross edge**. (2)



Provide an explanation for each classification, demonstrating how each type of edge occurs during the DFS traversal.

- 4) For the following graph, find the shortest path from A to E using Dijkstra's Algorithm: (5)

- A) **A → B = 3, A → C = 6, A → D = 1, B → C = 2, B → D = 4, B → E = 7, C → E = 1, C → F = 5, D → E = 3, D → F = 2, E → G = 6, F → G = 4, F → H = 3, G → H = 2, H → I = 1**

Show the steps involved in finding the shortest path from **A** to **E**.

- B) Create a singly linked list with the following values:
15, 25, 35, 45, 55, 65. (3)

- Insert a node at the beginning, at the end,
and at position 4.
- Display the final list after the operations.

Develop functions for the process of insertion.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Analyze the time complexity of the following code snippet using **Big O notation**: (2)

```
for(i = 0; i < n; i++)
for(j = 0; j < n; j++)
printf("%d", i + j);
```

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5) Given the infix expression: (5)
- (A + B) * (C - D) + (E / F)**

- A) • Construct an **expression tree**.

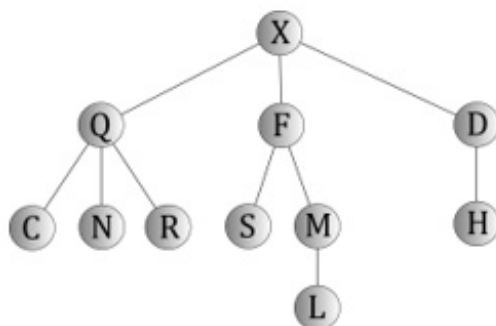
Perform **in-order, pre-order, and post-order** traversals of the tree.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Sort the following array using **Selection Sort**: (3)
- arr[10] = {85, 95, 75, 55, 65, 45, 35, 25, 15, 5}**
- Show the array after each pass.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Convert the following general tree to a binary tree: (2)



Draw the resulting binary tree

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5114:Data Structures and Algorithms

#	Course Outcome	Description	Marks
1	1	CO 1	11
2	2	CO 2	11

3	3	CO 3	9
4	4	CO 4	12
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5114:Data Structures and Algorithms

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)